



Features

- Proprietary α SiC™ MOSFET technology
- Low loss, fast switching speeds with low R_{DS}
- Optimized drive voltage ($V_{GS} = 15V$) for broad driver compatibility
- Robust body diode and low Q_{rr}

Product Summary

$V_{DS} @ T_{J, max}$	1200V
I_{DM}	120A
$R_{DS(ON), typ}$	33m Ω
Q_{rr}	226nC
$E_{OSS} @ 800V$	63 μ J
100% UIS Tested	

Applications

Renewable

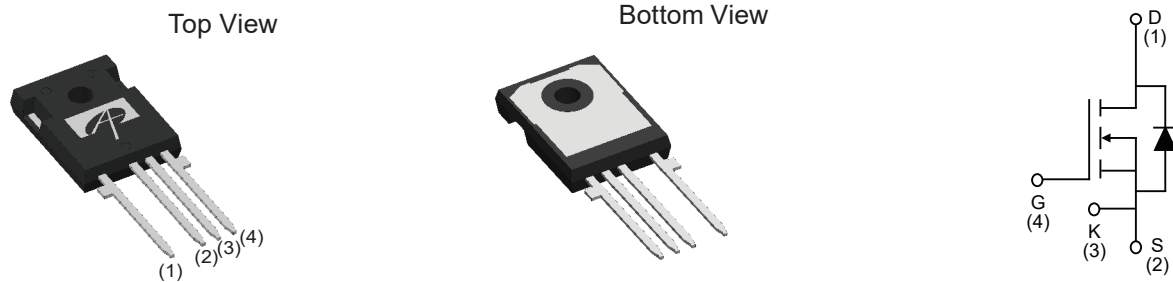
- EV Charger
- Solar Inverters

Industrial

- UPS
- SMPS
- Motor Drives



Pin Configuration



Ordering Part Number	Package Type	Form	Shipping Quantity
AOM033V120X2	TO-247-4L	Tube	30/Tube

Absolute Maximum Ratings

($T_A = 25^\circ C$, unless otherwise noted)

Symbol	Parameter		AOM033V120X2	Units
V_{DS}	Drain-Source Voltage		1200	V
$V_{GS, MAX}$	Gate-Source Voltage	Maximum	-8/+18	V
$V_{GS, OP, TRANS}$		Max Transient ^(A)	-8/+20	
$V_{GS, OP}$		Recommended Operating ^(B)	-5/+15	
I_D	Continuous Drain Current	$T_C = 25^\circ C$	68	A
		$T_C = 100^\circ C$	48	
I_{DM}	Pulsed Drain Current ^(C)		120	
E_{AS}	Single Pulsed Avalanche Energy ^(D)		1000	mJ
P_D	Power Dissipation ^(C)		300	W
T_J, T_{STG}	Junction and Storage Temperature Range		-55 to 175	$^\circ C$
T_L	Maximum lead temperature for soldering purpose, 1/8" from case for 5 seconds		300	$^\circ C$

Thermal Characteristics

Symbol	Parameter	AOM033V120X2	Units
$R_{\theta JA}$	Maximum Junction-to-Ambient ^(E,F)	40	°C/W
$R_{\theta JC}$	Maximum Junction-to-Case ^(G)	0.5	°C/W

Electrical Characteristics

($T_A = 25^\circ\text{C}$, unless otherwise noted)

Symbol	Parameter	Conditions	Min	Typ	Max	Units
STATIC PARAMETERS						
BV_{DSS}	Drain-Source Breakdown Voltage	$I_D = 250\mu\text{A}$, $V_{GS} = 0\text{V}$, $T_J = 25^\circ\text{C}$	1200			V
		$I_D = 250\mu\text{A}$, $V_{GS} = 0\text{V}$, $T_J = 150^\circ\text{C}$		1200		
I_{DSS}	Zero Gate Voltage Drain Current	$V_{DS} = 1200\text{V}$, $V_{GS} = 0\text{V}$, $T_J = 25^\circ\text{C}$			100	μA
I_{GSS}	Gate-Body Leakage Current	$V_{DS} = 0\text{V}$, $V_{GS} = +15/-5\text{V}$			± 100	nA
$V_{GS(th)}$	Gate Threshold Voltage	$V_{DS} = V_{GS}$, $I_D = 17.5\text{mA}$	2.3	2.8	3.5	V
$R_{DS(ON)}$	Static Drain-Source On-Resistance	$V_{GS} = 15\text{V}$, $I_D = 20\text{A}$		33	43	m Ω
		$T_J = 25^\circ\text{C}$				
		$T_J = 150^\circ\text{C}$		45		
g_{FS}	Forward Transconductance	$V_{DS} = 20\text{V}$, $I_D = 20\text{A}$		15	-	S
V_{SD}	Diode Forward Voltage	$I_S = 17.5\text{A}$, $V_{GS} = -5\text{V}$		4	5	V
DYNAMIC						
C_{iss}	Input Capacitance	$V_{GS} = 0\text{V}$, $V_{DS} = 800\text{V}$, $f = 1\text{MHz}$		2908		pF
C_{oss}	Output Capacitance			128		pF
C_{rss}	Reverse Transfer Capacitance			9.9		pF
E_{oss}	Coss Stored Energy			63		μJ
R_G	Gate Resistance	$f = 1\text{MHz}$		1.7		Ω
SWITCHING						
Q_g	Total Gate Charge	$V_{GS} = 5/+15\text{V}$, $V_{DS} = 800\text{V}$, $I_D = 20\text{A}$, $I_{GS} = 50\text{mA}$		104		nC
Q_{gs}	Gate Source Charge			37		nC
Q_{gd}	Gate Drain Charge			32		nC
$t_{d(on)}$	Turn-On Delay Time	$V_{GS} = -5\text{V}/+15\text{V}$, $V_{DS} = 800\text{V}$, $I_D = 40\text{A}$, $R_G = 2\Omega$		13.3		ns
t_r	Turn-On Rise Time			15.4		ns
$t_{d(off)}$	Turn-Off Delay Time			15.3		ns
t_f	Turn-Off Fall Time			5.1		ns
E_{on}	Turn-On Energy			297		μJ
E_{off}	Turn-Off Energy	FWD: AOM033V120X2		40		μJ
E_{tot}	Total Switching Energy			337		μJ
t_{rr}	Body Diode Reverse Recovery Time	$I_F = 20\text{A}$, $dI/dt = 1500\text{A}/\mu\text{s}$, $V_{GS} = -5\text{V}$, $V_{DS} = 800\text{V}$		48		ns
I_{rm}	Peak Reverse Recovery Current			13		A
Q_{rr}	Body Diode Reverse Recovery Charge			226		nC

Notes:

- $t_{pulse} < 1\mu\text{s}$, $f > 1\text{Hz}$
- Device can be operated at $V_{GS} = 0/15\text{V}$. Actual operating V_{GS} will depend on application specifics such as parasitic inductance and dV/dt but should not exceed maximum ratings.
- The power dissipation P_D is based on $T_{J(MAX)} = 175^\circ\text{C}$, using junction-to-case thermal resistance, and is more useful in setting the upper dissipation limit for cases where additional heatsinking is used.
- $L = 5\text{mH}$, $I_{AS} = 20\text{A}$, $R_G = 25\Omega$. Starting $T_J = 25^\circ\text{C}$.
- The value of $R_{\theta JA}$ is measured with the device in a still air environment with $T_A = 25^\circ\text{C}$.
- The $R_{\theta JA}$ is the sum of the thermal impedance from junction to case $R_{\theta JC}$ and case to ambient.
- The value of $R_{\theta JC}$ is measured with the device mounted to a large heatsink, assuming a maximum junction temperature of $T_{J(MAX)} = 175^\circ\text{C}$.
- The static characteristics in Figures 1 to 8 are obtained using $< 300\text{ms}$ pulses, duty cycle 0.5% max.
- These curves are based on $R_{\theta JC}$ which is measured with the device mounted to a large heatsink, assuming a maximum junction temperature of $T_{J(MAX)} = 175^\circ\text{C}$. The SOA curve provides a single pulse rating.

Typical Electrical and Thermal Characteristics

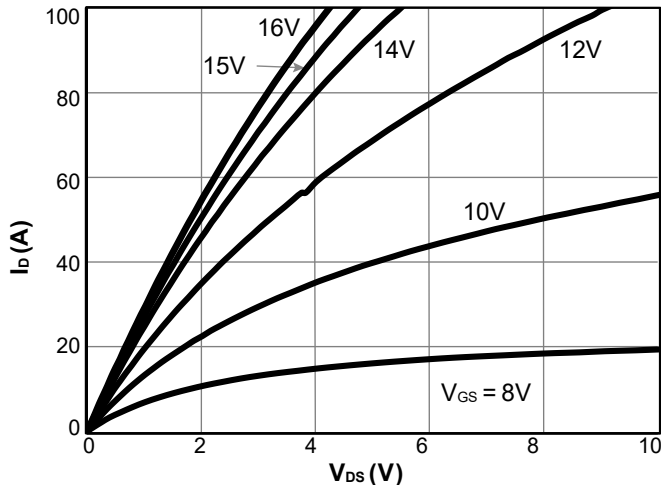


Figure 1. On-Region Characteristics $T_j = 25^\circ\text{C}$

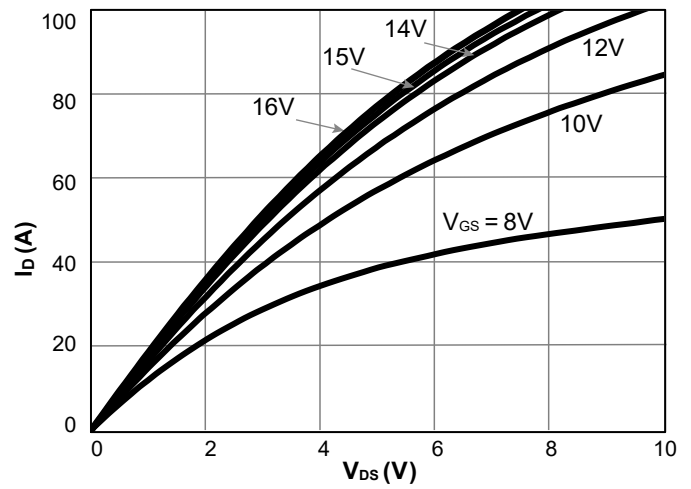


Figure 2. On-Region Characteristics $T_j = 175^\circ\text{C}$

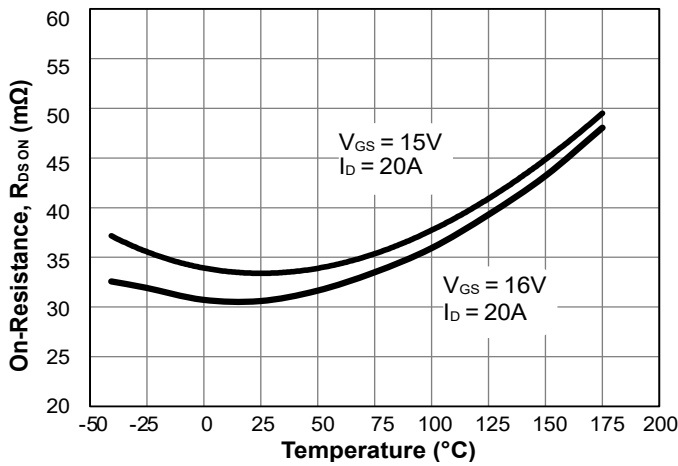


Figure 3. On-Resistance vs. Junction Temperature

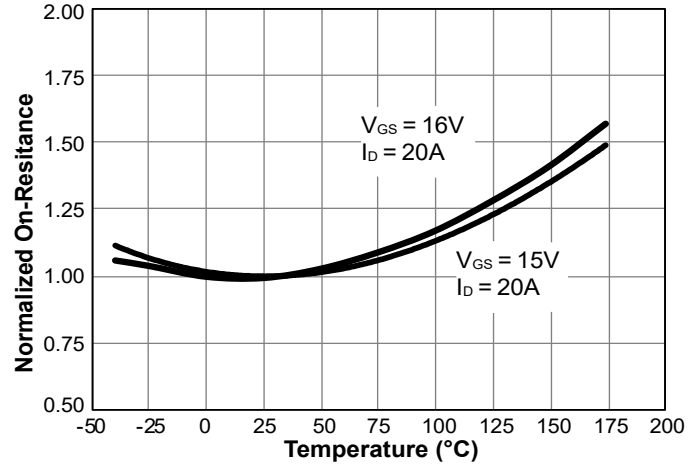


Figure 4. Normalized On-Resistance vs. Junction Temperature

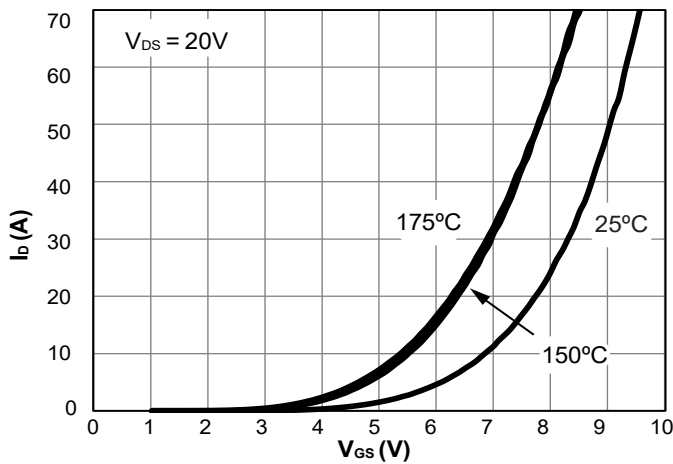


Figure 5. Transfer Characteristics

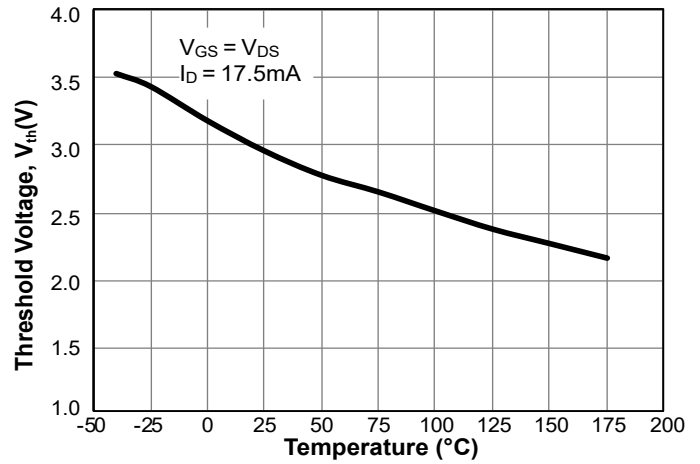


Figure 6. Threshold Voltage vs. Junction Temperature

Typical Electrical and Thermal Characteristics (Continued)

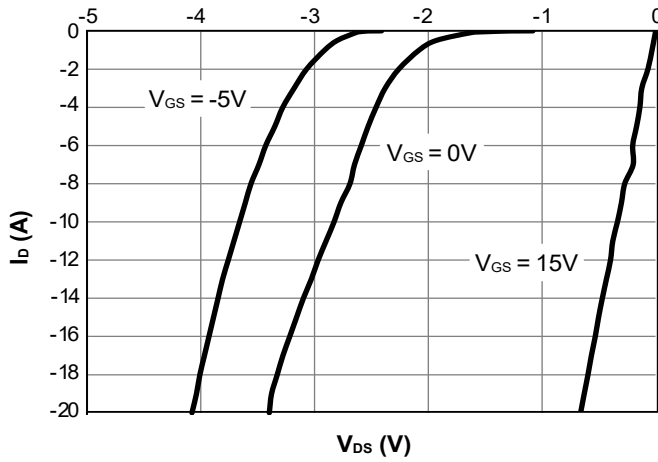


Figure 7. Body-Diode Characteristics at 25°C

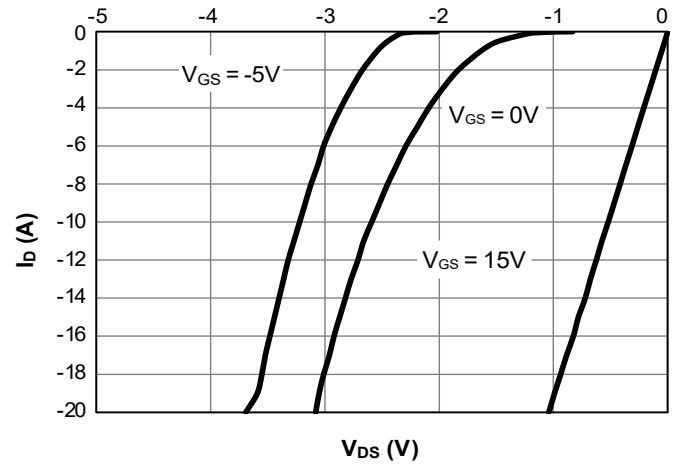


Figure 8. Body-Diode Characteristics at 175°C

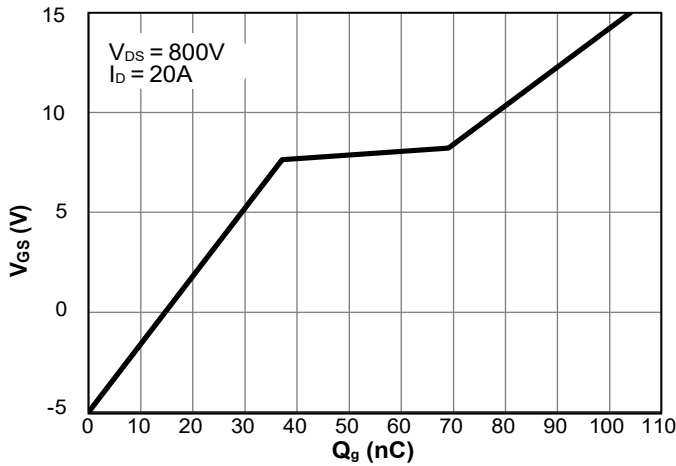


Figure 9. Gate-Charge Characteristics

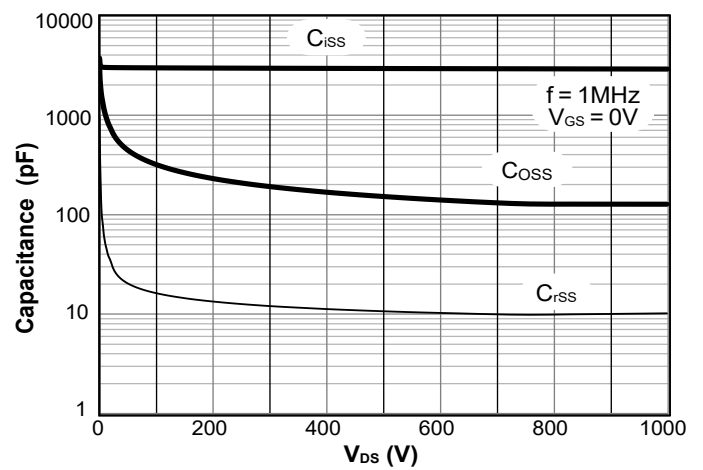


Figure 10. Capacitance Characteristics

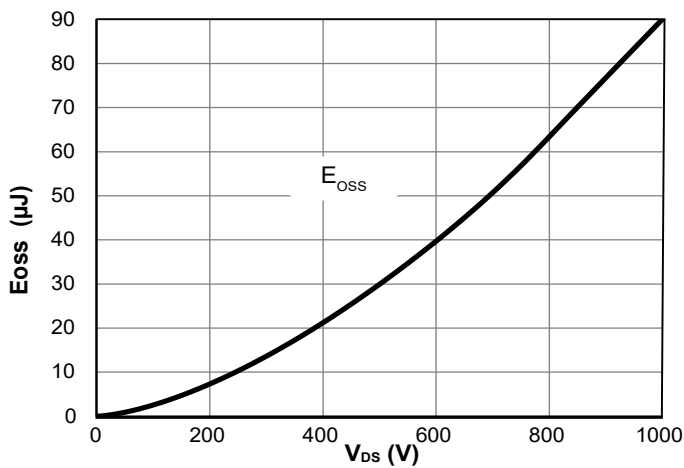


Figure 11. Coss Stored Energy

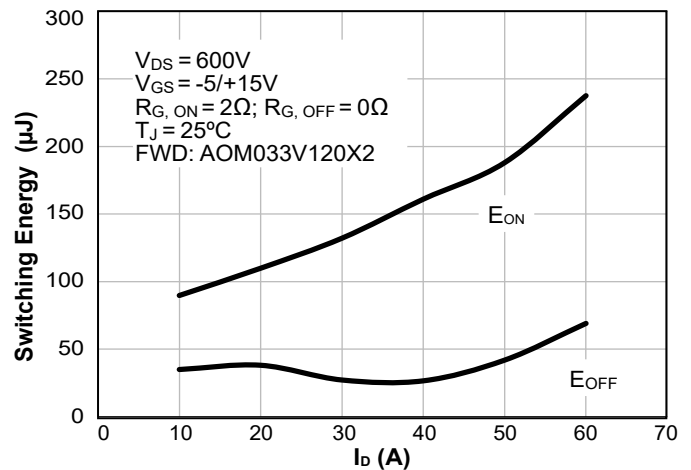


Figure 12. Switching Energy vs. Drain Current

Typical Electrical and Thermal Characteristics (Continued)

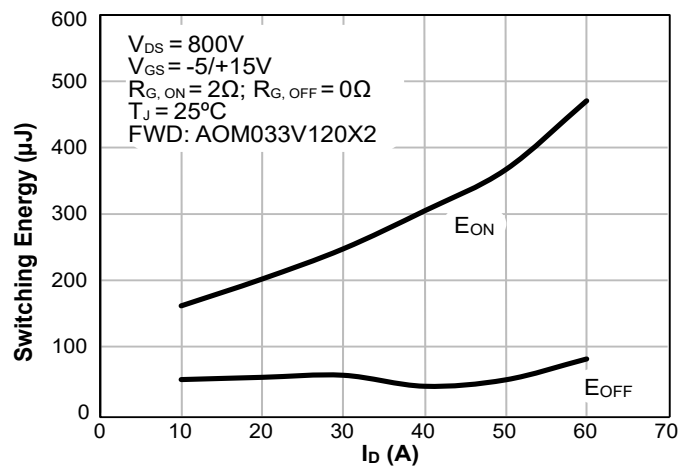


Figure 13. Switching Energy vs. Drain Current

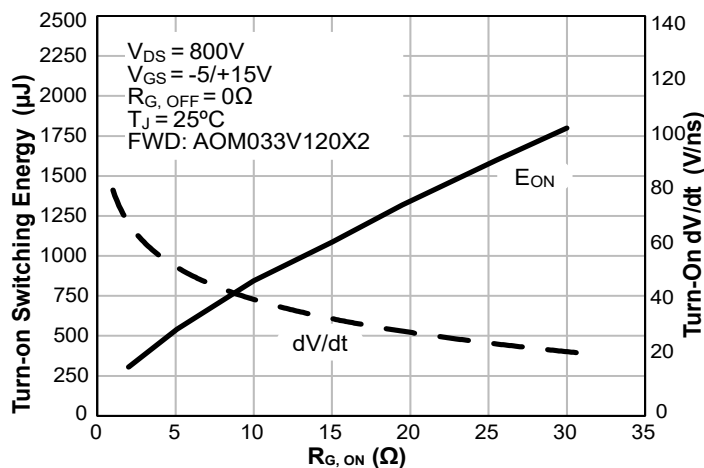


Figure 14. Turn-On Energy and dV/dt vs. External Gate Resistance

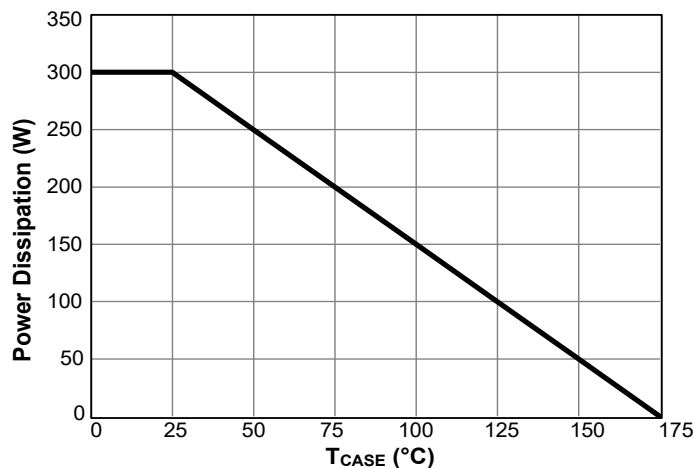


Figure 15. Power Derating vs. Case Temperature (Note I)

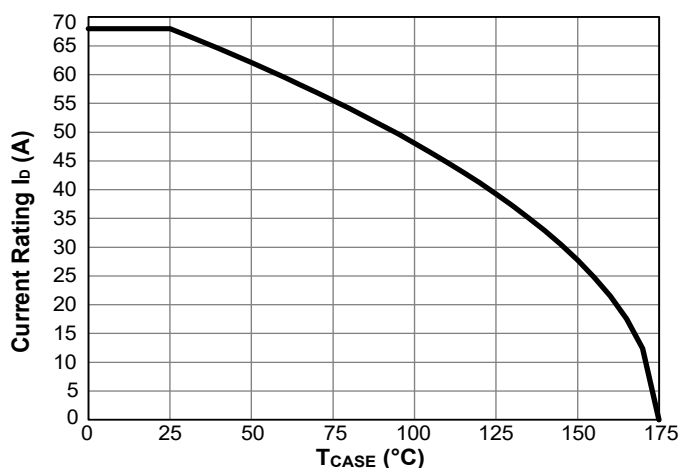


Figure 16. Current Derating vs. Case Temperature (Note I)

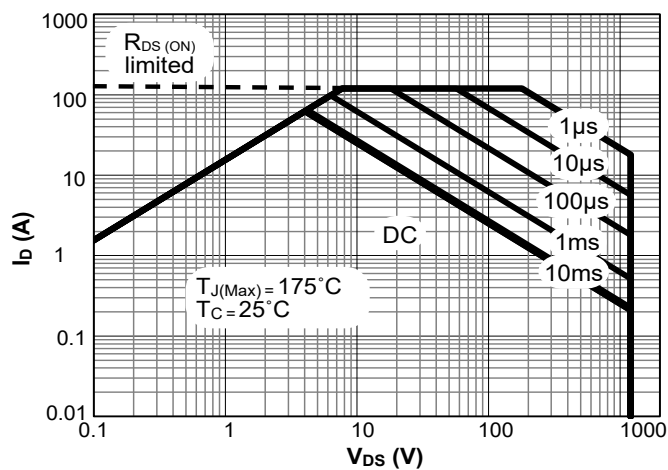


Figure 17. Maximum Forward Biased Safe Operating (Note I)

Typical Electrical and Thermal Characteristics (Continued)

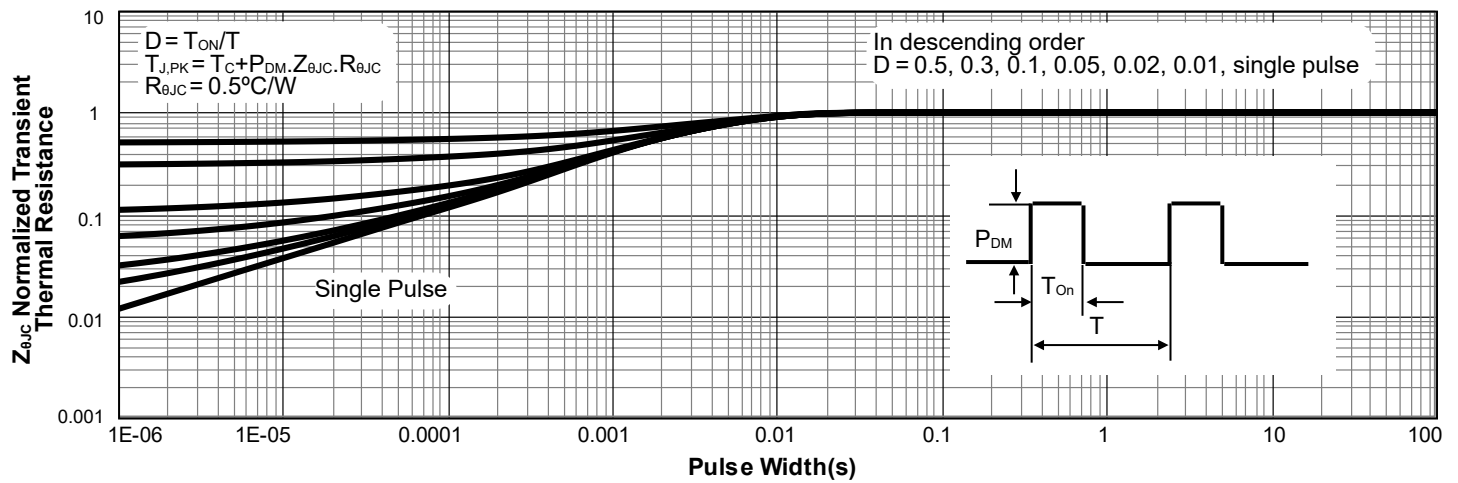


Figure 18. Normalized Maximum Transient Thermal Impedance for AOM033V120X2 (Note I)

Test Circuits and Waveforms

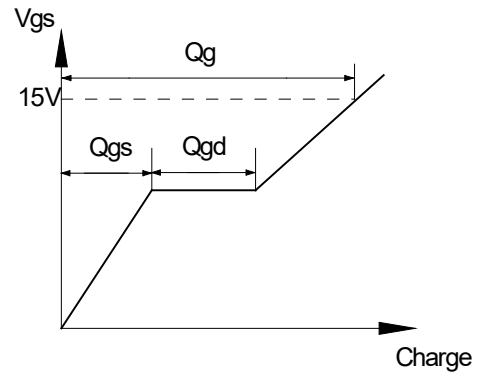
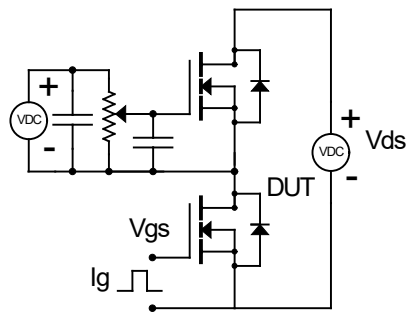


Figure 19. Gate Charge Test Circuits and Waveforms

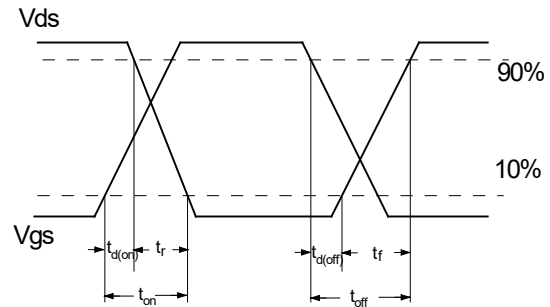
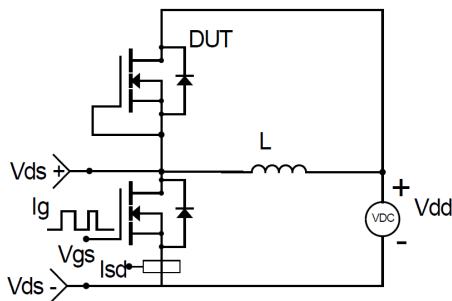


Figure 20. Inductive Switching Test Circuit and Waveforms

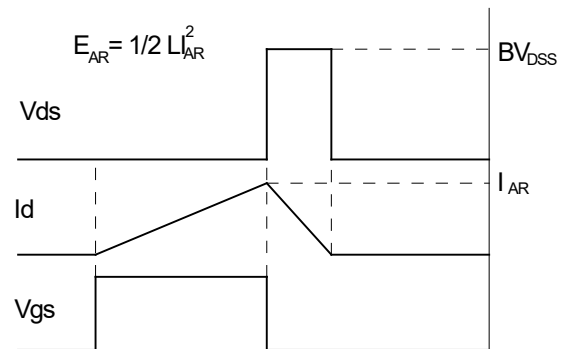
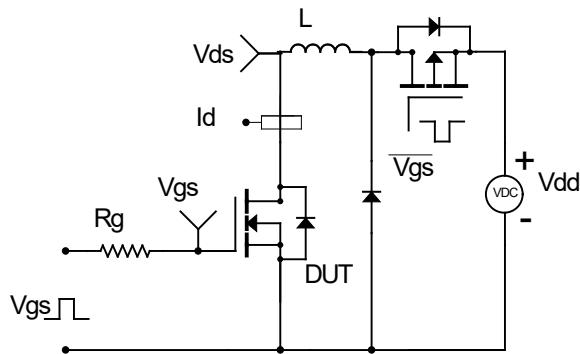


Figure 21. Unclamped Inductive Switching (UIS) Test Circuit and Waveforms

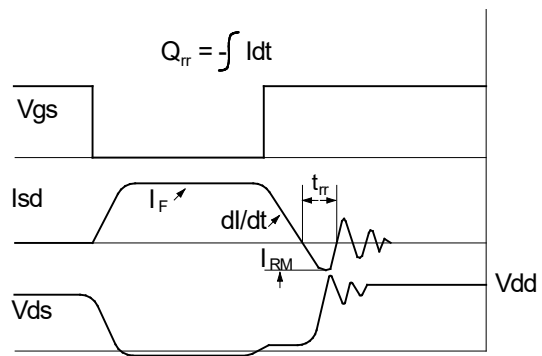
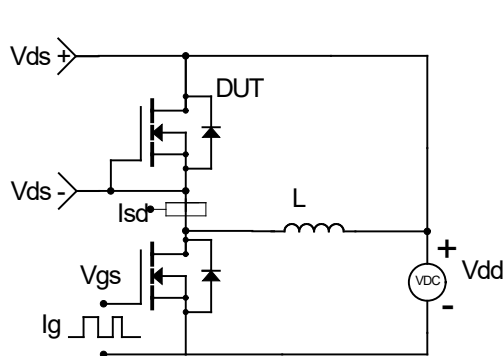
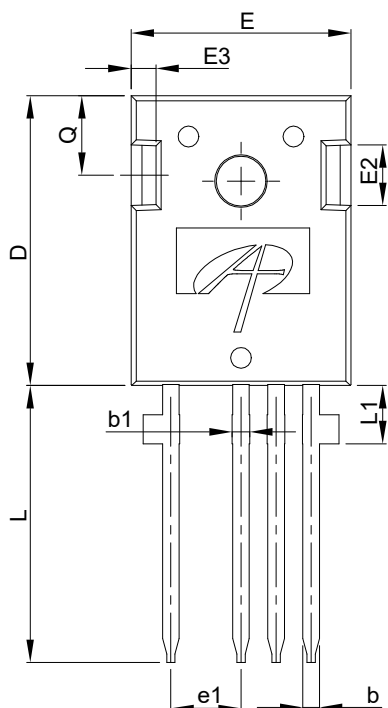
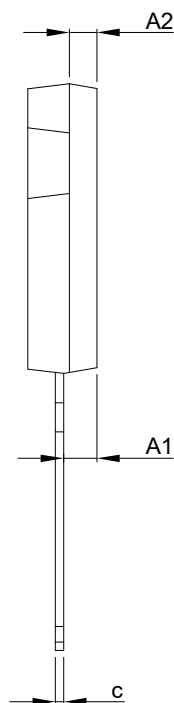


Figure 22. Diode Recovery Test Circuits and Waveforms

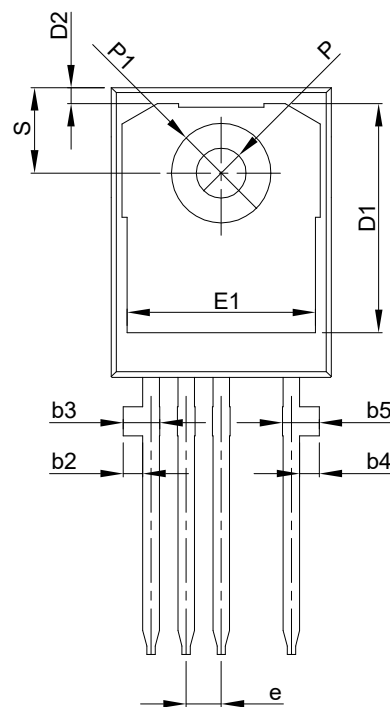
Package Dimensions, TO-247-4L



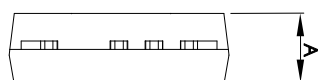
TOP VIEW



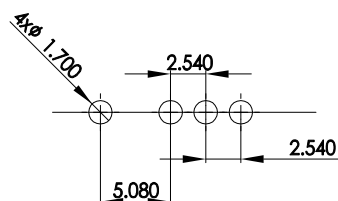
SIDE VIEW



BOTTOM VIEW



SIDE VIEW



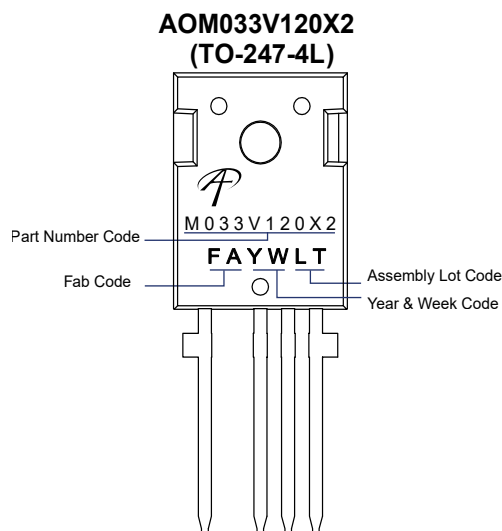
RECOMMENDED THROUGH HOLES
FOR LAND PATTERN

NOTE:

1. CONTROLLED DIMENSIONS ARE IN MILLIMETERS.
2. THIS IS AN ENGINEERING DRAFT FOR REVIEW. AOS CONFIDENTIAL.

SYMBOLS	DIM. IN MM			DIM. IN INCH		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	4.90	5.00	5.10	0.193	0.197	0.201
A1	2.32	2.42	2.52	0.091	0.095	0.099
A2	1.90	2.00	2.10	0.075	0.079	0.083
b	1.17	1.22	1.27	0.046	0.048	0.050
b1	1.20	1.30	1.40	0.047	0.051	0.055
b2	1.31	1.41	1.51	0.052	0.056	0.059
b3	2.45	2.65	2.85	0.096	0.104	0.112
b4	1.31	1.41	1.51	0.052	0.056	0.059
b5	2.45	2.65	2.85	0.096	0.104	0.112
c	0.57	0.62	0.67	0.022	0.024	0.026
D	20.80	20.95	21.10	0.819	0.825	0.831
D1	16.25	16.55	16.85	0.640	0.652	0.663
D2	1.00	1.15	1.30	0.039	0.045	0.051
E	15.77	15.92	16.07	0.621	0.627	0.632
E1	13.43	13.63	13.83	0.529	0.536	0.544
E2	4.29	4.39	4.49	0.169	0.173	0.177
E3	1.70	1.80	1.90	0.067	0.071	0.075
e	2.54BSC			0.1000BSC		
e1	5.08BSC			0.2000BSC		
N	4			4		
L	19.82	20.02	20.22	0.780	0.788	0.796
L1	4.01	4.21	4.41	0.158	0.166	0.174
P	3.50	3.60	3.70	0.138	0.142	0.146
P1	7.00	7.20	7.40	0.276	0.283	0.291
Q	5.65	5.75	5.85	0.222	0.226	0.230
S	6.07	6.17	6.27	0.239	0.243	0.247

Part Marking



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2. A critical component in any component of a life support, device, or system whose failure to perform can be reasonably expected to cause the failure of the life support device or system, or to affect its safety or effectiveness.